

Financial KPI Analysis Using Walmart Sales Data

1. Introduction

The objective of this project is to analyze Walmart's weekly sales data and evaluate key financial performance indicators (KPIs). Businesses use KPI analysis to monitor sales performance, identify trends, compare store performance, and support data-driven decision-making.

In this project, Python was used for data cleaning, preprocessing, and exploratory data analysis (EDA). Tableau was used to build an interactive dashboard that visualizes sales trends, store performance, holiday impact, and yearly sales analysis.

The project demonstrates the complete data analysis workflow, including data preparation, visualization, and business insight generation.

2. Objectives

The main objectives of this project are:

- Analyze Walmart weekly sales data.
 - Clean and preprocess the dataset.
 - Calculate important financial KPIs.
 - Identify the highest-performing stores.
 - Analyze monthly and yearly sales trends.
 - Compare holiday and non-holiday sales.
 - Build an interactive Tableau dashboard.
 - Generate business insights to support decision-making.
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3. Dataset Description

The dataset contains Walmart weekly sales information collected from multiple stores over different weeks.

Dataset Features

- Store
- Date
- Weekly_Sales

- Holiday_Flag
- Temperature
- Fuel_Price
- CPI
- Unemployment
- Year
- Month
- Month_Name

The dataset contains sales records from multiple Walmart stores and includes economic indicators that may influence sales performance.

4. Tools and Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Tableau
 - Jupyter Notebook
 - Visual Studio Code
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5. Methodology

The project was completed using the following steps:

Data Collection

The Walmart sales dataset was downloaded and imported into Python for analysis.

Data Cleaning

The following preprocessing tasks were performed:

- Checked missing values.

- Checked duplicate records.
- Converted the Date column into datetime format.
- Created Year, Month, and Month_Name columns.
- Verified data quality.

Exploratory Data Analysis (EDA)

The dataset was analyzed to identify:

- Total weekly sales.
- Average weekly sales.
- Highest and lowest sales.
- Top-performing stores.
- Monthly sales trends.
- Yearly sales trends.
- Holiday sales comparison.
- Relationship between sales and economic factors.

Dashboard Development

A Tableau dashboard was created to visualize important KPIs and business insights.

6. Key Performance Indicators (KPIs)

The following KPIs were calculated:

- Total Sales
- Average Weekly Sales
- Highest Weekly Sales
- Lowest Weekly Sales
- Total Number of Stores
- Best Performing Store
- Best Performing Month
- Holiday Sales
- Non-Holiday Sales

These KPIs help management evaluate business performance efficiently.

7. Dashboard Features

The Tableau dashboard contains the following visualizations:

- Monthly Sales Trend
- Sales by Year
- Top 10 Stores by Sales
- Holiday vs Non-Holiday Sales
- Sales Distribution
- Store Performance Heatmap

The dashboard also supports interactive filtering to improve data exploration.

8. Business Insights

The analysis provided several valuable insights:

- Some Walmart stores consistently generate significantly higher sales than others.
 - Monthly sales fluctuate throughout the year, indicating seasonal demand.
 - Holiday periods influence customer purchasing behavior.
 - Sales performance varies across different years.
 - Economic factors such as fuel price, CPI, and unemployment have a relatively weak relationship with weekly sales.
 - Tableau dashboards allow managers to monitor KPIs more effectively.
 - Identifying top-performing stores helps management understand successful business strategies.
 - Seasonal sales analysis supports better inventory planning.
 - KPI monitoring enables faster business decision-making.
 - Data visualization simplifies the interpretation of large datasets.
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9. Conclusion

This project successfully analyzed Walmart sales data using Python and Tableau. Data cleaning, preprocessing, KPI calculation, and visualization were performed to understand sales performance across stores and time periods.

The interactive Tableau dashboard provides an effective way to monitor financial KPIs and supports business decision-making through clear visualizations. This project demonstrates practical data analysis skills, including data preprocessing, exploratory data analysis, dashboard creation, and business insight generation.

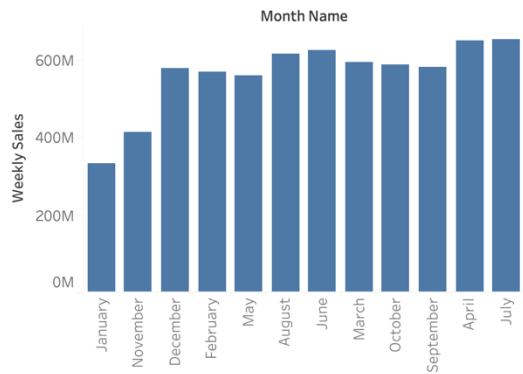
10. Future Scope

Future improvements to this project include:

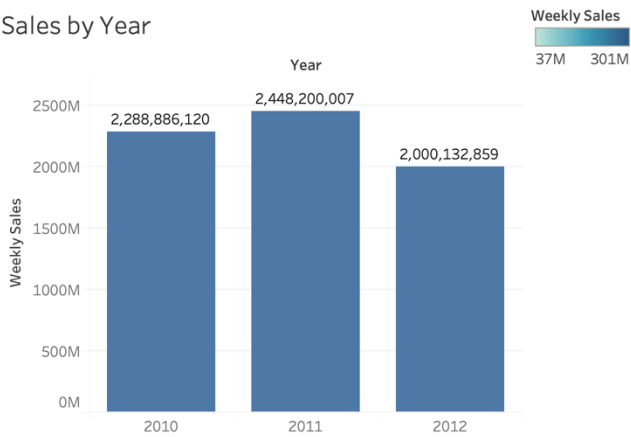
- Building a sales forecasting model using machine learning.
- Predicting future weekly sales.
- Integrating real-time sales data.
- Developing a web-based analytics dashboard.
- Adding customer segmentation analysis.
- Creating automated KPI reporting.

Walmart Financial KPI Dashboard

Monthly Sales Trend



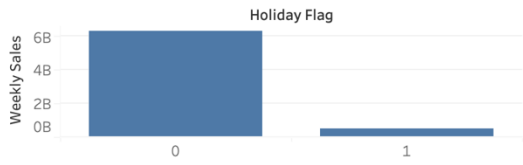
Sales by Year



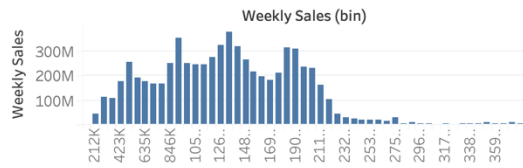
Top 10 Stores

Store	
1	Abc
2	Abc
4	Abc
6	Abc
10	Abc
13	Abc

Holiday vs Non-Holiday Sales



Sales Distribution



Store Performance

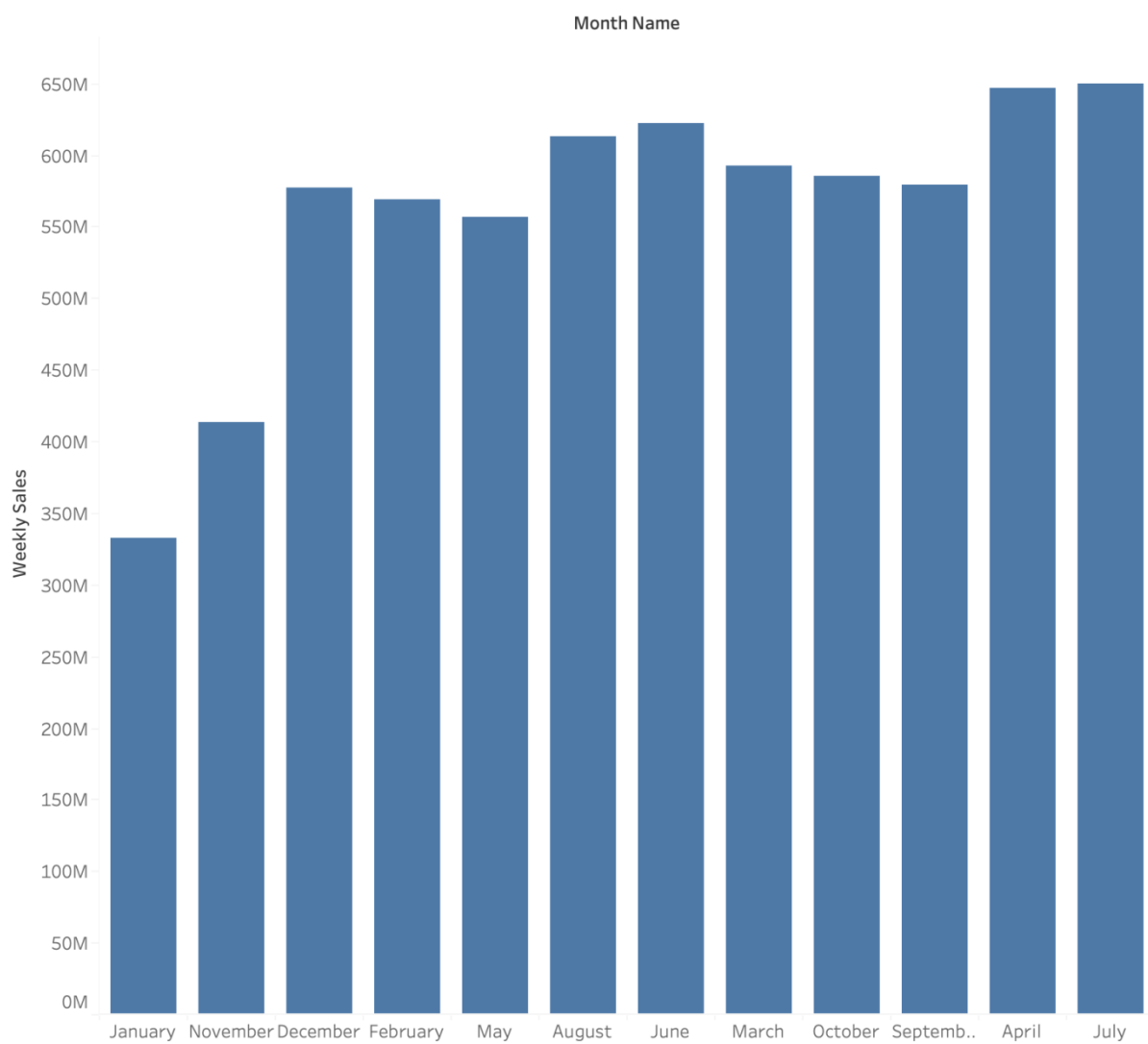
Store	
1	222,402,809
2	275,382,441
3	57,586,735
4	299,543,953

Top 10 Stores

Store	
1	Abc
2	Abc
4	Abc
6	Abc
10	Abc
13	Abc
14	Abc
20	Abc
27	Abc
39	Abc

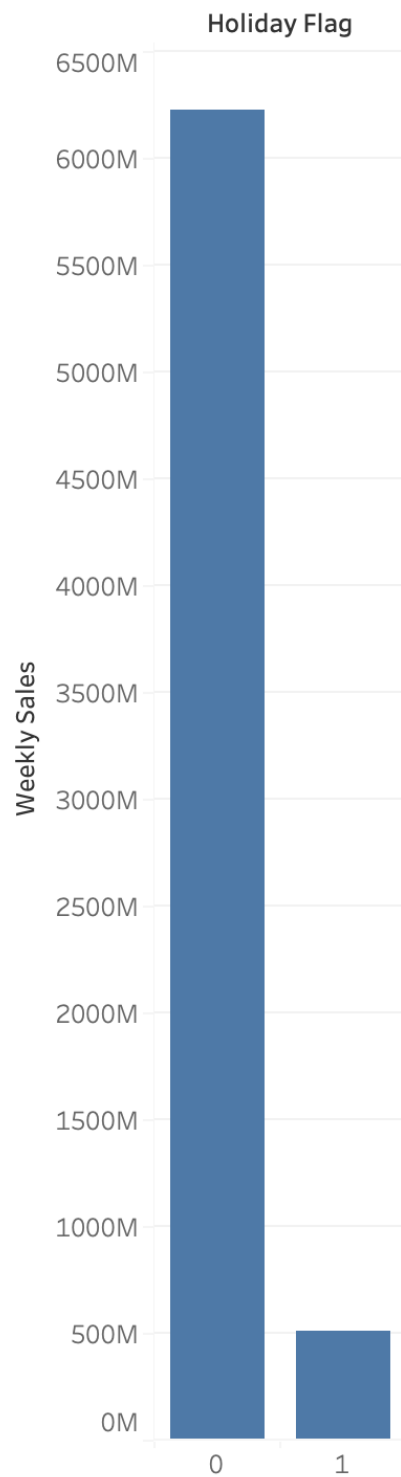
The view is broken down by Store. The view is filtered on Store, which keeps 10 of 45 members.

Monthly Sales Trend



Sum of Weekly Sales for each Month Name. The view is filtered on Month Name, which keeps 12 of 12 members.

Holiday vs Non-Holiday Sales



Sum of Weekly Sales for each Holiday Flag. The view is filtered on Holiday Flag and sum of Weekly Sales. The Holiday Flag filter keeps 0 and 1. The sum of Weekly Sales filter includes everything.